

An Enhanced Security Protocol for Smart Homes

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Abstract—Over the past several years, home automation systems have become the target of cyberattacks due to a lack of efficient security mechanisms for low-power devices. These attacks made the requirement for a security framework or architecture that is both efficient and lightweight. To make the protocol resilient to attacks like phishing or spoofing, this research includes the use of biometrics and sensor fingerprinting, which is partially validated using machine learning. Both biometrics and sensor fingerprinting are susceptible to hacking when used in their typical application, but combining those two and simulating the way of multifactor authentication can increase the resiliency. In this project, we intend to implement a machine learning algorithm in the protocol flow, which does liveness detection on fingerprints provided and another model to identify the sensor based on the raw accelerometer data provided. We propose that these validations can mitigate attacks like spoofing, and the protocol in its entirety prevents replay attacks and many others while still being lightweight. We expect this framework to be utilised on a variety of devices, which can be a major step considering the use of machine learning in the field of the Internet of Things for authentication.

Index Terms—Machine Learning, Sensor Fingerprinting, Authentication Protocol, Internet of Things, Geometric Secret Sharing, Finger Biometrics

I. INTRODUCTION

The rise in attacks against smart home systems is largely related to the lack of security frameworks that are robust and efficient. Analysis done in past years shows a substantial gain of 124% in the past year and nearly up to 400% cumulatively for the past years. Moreover, it also shows an average of 10 attacks on smart homes per day. This shows the need for having security measures or a framework designed which are specifically suitable for smart homes.

Research done in the past years shows significant advancements, yet a comprehensive framework that meets all the requirements to provide adequate security is very few to none. Nimmy et al. in [16] present a framework for mutual authentication for IoT-based devices. On performing security and performance assessments, they show their superiority when compared with few other frameworks. In the domain of biometrics, machine learning techniques for sensor and fingerprint analysis have been explored. Berdich et al. [17], Hupperich et al. [8], Bojinov et al. [5], and Ding and Ming [12] provide impactful contributions. Similarly, we cite Chowdhury et al. [15], Nogueira et al. [9], Praseetha et al. [13], and Kothadiya

et al. [18]. They employ different deep learning or machine learning models, making it efficient and robust to outliers.

The existing systems have notable drawbacks, mostly having security vulnerabilities or having the need for high computational resources. The primary aim of this study is to provide the solution which is lightweight and resilient. We aim to utilise machine learning models which are trained and incorporate those in the framework developed by Nimmy et al., as we believe there can be significant improvement done by replacing the use of Photo Response Non Uniformity (PRNU), which is done for device identification, with sensor fingerprinting and face recognition with biometrics, along with liveness detection to prevent spoofing attacks. The above changes are considered because PRNU is easily affected by image processing and spatial synchronisation issues, and facial recognition on low-power devices can be resource-intensive.

This study employs a framework established by Nimmy et al. [16] and implements essential modifications to enhance the existing framework while creating a framework that can leverage machine learning to be more accurate.

A. Our Contributions

These include

- 1) Sensor fingerprinting by the use of machine learning model instead of PRNU
- 2) finger biometrics along with liveness detection, where liveness detection is done using a machine learning model

We will be training machine learning models which use fewer resources for inferences. Our expectation is to have robustness and resiliency while keeping the framework lightweight. We know that training the model on the right data can make the model very accurate by reducing the false positives while being resilient to many known attacks like spoofing, etc. The intention of using machine learning in the IoT domain specifically for mutual authentication can be a positive impact due to its early research stage. For the purpose of this project we expect to train a model for liveness detection and one for sensor fingerprinting.

II. RELATED WORK

As previously stated, we are utilising the framework established by Nimmy et al. [16] as the foundation of our

research and modifying the existing framework to enhance its robustness. The key elements of the framework include the integration of PRNU, facial recognition for the identification of devices and users. In addition to the mentioned methods, they employ Elliptic Curve Cryptography for the generation of public and private keys, utilise symmetric key algorithms, and implement basic XOR and hashing techniques to scramble data for privacy, ensuring that only authorised users may access the information. Numerous studies have explored sensor fingerprinting for accelerometers, utilising both machine learning and non-machine learning techniques. Berdich et al. [17] propose the combination of traditional and CNN approaches to get enhanced accuracy and robust results. Hupperich et al. [8] investigate temporal and frequency domain analysis for feature extraction employed in fingerprinting. Bojinov et al. [5] demonstrate entropy sufficient for the distinct identification of devices, which remains robust against substantial modifications. Ding and Ming [12] leverage the use of traditional classifiers along with deep learning, along with feature extraction and anomaly detection for higher accuracy and lightweight computation for mobile environments. All the research cited about sensor fingerprinting functions effectively; nevertheless, they are either resource-intensive, need big datasets, or demand rigorous feature selection. A study by Chowdhury and Imtiaz [15] on finger biometrics indicates the necessity for structured light, high-resolution cameras, and sophisticated image algorithms. They have employed deep learning for image preprocessing and enhancement. Nogueira et al. [9] advocate for the utilisation of convolutional neural networks (CNNs) owing to their exceptional accuracy, even with limited datasets. Kothadiya et al. [18] use ResNet and data augmentation, resulting in enhanced resilience and accuracy. Praseetha et al. [13] use deep learning and a minutiae-based approach to enhance security and minimise false positives. Like the limitations of the mentioned sensor fingerprinting, we also need a proper dataset which is balanced to avoid very common issues like generalisation and the resource requirement when considering the use of sophisticated deep learning models.

III. BACKGROUND

A. SMARTPHONE SENSOR IDENTIFICATION

The accelerometer is one of the basic sensors present in today's smartphones due to their extensive use, from basic screen orientation to navigation support. There are considerable amounts of research and implementation of sensor fingerprinting both with and without the use of machine learning. This fingerprinting method uses the raw time series data and recognises the pattern in it to uniquely identify the device. To provide the model with more sophisticated data, we plan to extract time and frequency domain features and use them for the training. Extracted data is then sent to the model for training which makes the model suitable for identification.

B. BIOMETRIC IDENTIFICATION

We leverage matching the fingerprint of the user to an existing fingerprint. Though it is simple, it also comes along

with its security concerns, like spoofing, which they are easily susceptible to. For fingerprint matching we use Scale-Invariant Feature Transformation (SIFT) along with Random Sample Consensus (RANSAC).

1) **SIFT**: It is an algorithm developed to describe and detect local features in an image. The method is resilient to changes in the scale of the image brightness and also the orientation of the image. We apply a Gaussian filter to the provided image to pinpoint crucial points such as corners and edges. Once the points are identified, they are pinpointed, and points along the edges are discarded. Based on the local image gradient directions, we assign one or more orientations to each point. The descriptor encodes the patch of the pixel around the location and is stored as a vector, after which it is matched across different images to identify.

2) **RANSAC**: RANSAC is a non-deterministic algorithm that helps identify the false matches done by SIFT. The algorithm identifies the number of data points that are consistent. This is robust to outliers and capable of providing reliable estimates.

C. MUTUAL AUTHENTICATION

Mutual authentication is achieved using geometric secret sharing; a secret is divided into shares and allocated among participants, ensuring that only authorised individuals can reconstruct the secret. Shamir [1], Blakley, and Kabatianskii [3] proposed a threshold secret sharing technique that provides shares among n participants, requiring t participants for reconstruction. The fundamental concept is that two points are adequate to construct a line, with the key being the coordinates of a constant point on a specified line that intersects the y-axis at a secret point [14]. When having a random slope and two points, we can deduce a line l with y-intercept as a secret. You can generate the shares, as shown in 1, where P is a prime number and the shares can be reconstructed as shown in 2

$$\begin{aligned} share_1 &= (Secret + RandomSlope) \bmod P \\ share_2 &= (Secret + 2RandomSlope) \bmod P \end{aligned} \quad (1)$$

$$Secret = (2 * share_1 - share_2) \bmod P \quad (2)$$

IV. PROPOSED MODEL TRAINING

Since the integration of machine learning might make the model resource-intensive, we plan on using models which might take the least possible amount of resources to keep the protocol lightweight.

A. LIVENESS DETECTION

For this, we propose the use of the Naive Bayes model due to its features, like low computational complexity, low memory usage, ideal usage with high-dimensional data, and simplicity of implementation.

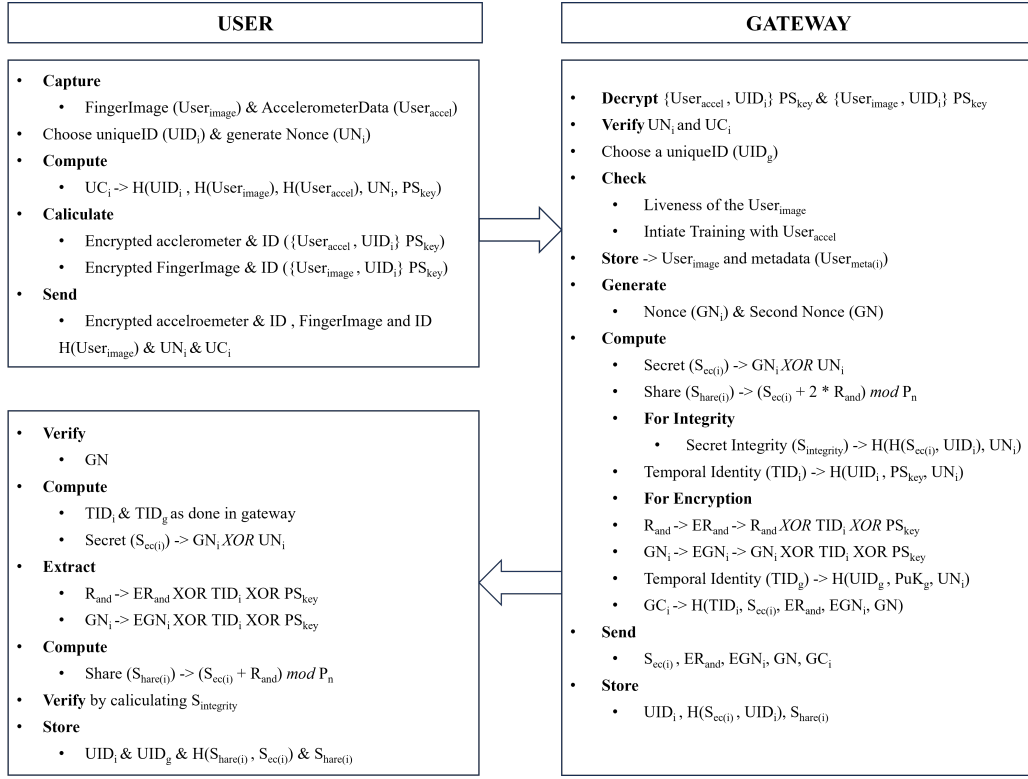


Fig. 1. User Registration in Gateway

1) **TRAINING**: For training the model on live and spoof images, we have leveraged the dataset provided by [10]. We have trained a machine learning model and a few deep learning models like NASNetMobile and MobileNetV3, which are specific for low-resource devices. We choose Naive Bayes, as the training time is almost the same or less than a deep learning model while providing the same accuracy using fewer resources. The flow of the training includes feature extraction, which is done using Local Binary Pattern (LBP), which is sent into the model, and the model is trained on tuning var_smoothing value which is added to each feature, which helps avoid numerical instability after which the model which has best accuracy is chosen.

B. SENSOR FINGERPRINTING

Similarly, for training machine learning models, we have used a combination of multiclass random forest and single-class SVM. Random forest helps to identify the device by taking the raw accelerometer data as input, from which multiple time and frequency domain features are extracted and sent to the model for prediction. SVM is trained to identify if the device is known or unknown. We are using a process similar to [6]. We have collected data from 7 different devices using Matlab Mobile, which has an inbuilt sensor raw data capturing option. The flow includes the detection of a flat static state; if found, then the static data is filtered out; else the raw data is considered. Feature extraction of multiple time and frequency domains is performed on the raw data, which is

Notation	Description
$PuK_{user/gateway/device}$	Public key of user, gateway or device
$PrK_{user/gateway/device}$	private key of user, gateway or device
H	Hash function
Gen	Generator
P_n	Prime Number
PS_{key}	Pre-Shared Keys
UN_i, GN, UN_j	User/Gateway/Device generated nonce
$User_{image}, User_{accel}$	Biometric image, raw accelerometer data
R_{and}	Random Number
TID_x	Temporal Identity of user / device
UID_x	Unique Identity of user / device
$Sec_{i/j}$	Secret of user/device
$Share_{i/j}$	Share of user/device
S_{key}	Session Key
$RREG$	Re-Registration

TABLE I

NOTATIONS USED IN PROPOSED PROTOCOL

then normalised and split for training and testing of the model, using which a random forest and a one-class SVM are trained.

V. PROPOSED PROTOCOL

Nimmy et al. proposed a hybrid key-based protocol, using asymmetric keys for generating temporal identities; hash and XOR are used for authentication, key agreement. To this protocol we propose the integration of finger biometrics and sensor fingerprinting, which is done using machine learning while keeping it as lightweight as possible.

In this section we discuss the proposed authentication protocol for smart homes using geometric secret sharing. This pro-

toocol, like the existing one, has four phases: the initialisation phase, the device or user registration phase, the authentication and key establishment phase and the re-registration phase.

A. THREAT MODEL

1) HOME NETWORK:

In a home network the entities in question are users, smart devices, and gateways. To access devices on the home network, a user must first authenticate with the gateway. A smart home network typically consists of n users, identified as users $User_i$. with i ranging from 1 to n , m smart gadgets labeled as $Device_j$ with j ranging between (1, m) and a gateway as $Gateway$. The devices are connected via the gateway. User registration and authentication for smart devices and users is handled by the gateway. All queries are sent via the gateway, which authenticates them prior to transmission. For connecting to device $Device_j$ on the network, the request is sent to the $Gateway$ by the user $User_i$, after which $Gateway$ verifies the authentication of user $User_i$ and sends messages to both $User_i$ and $Device_j$. These messages aid in the authentication of the gateway to the user and device. Using this solution, only an authenticated user $User_i$ and a device $Device_j$ can compute the session key and communicate with each other.

2) THREAT MODEL:

Usage of the Dolev-Yao threat model [2] for the protocol is best suited. This includes

- 1) user or smart device cannot be trusted
- 2) adversary is assumed to have access to the network and the entire traffic. He can
 - a) actively eavesdrop to obtain the message
 - b) send, delete or modify the message
 - c) replay the message
- 3) For testing this protocol, we can consider threats like
 - a) Installation of fake gateway
 - b) duplicate smart device
 - c) adversary can capture the smart device and perform attacks to extract information

3) ASSUMPTIONS:

For the case of testing or developing the framework, we assume,

- 1) Devices are added to the gateway in offline mode
- 2) During registration phase, the gateway is completely trusted
- 3) Secure key Storage is used for storing the keys in the device
- 4) Users smartphone is secured

4) INITIALIZATION PHASE:

We generate public and private keys using Elliptic Curve Cryptography (ECC). Gateway for this selects an elliptic curve $E(Field_p)$ over a finite field $Field_p$ which is defined by the equation $y^2 = x^3 + ax + b \mod Pn$ with a generator Gen where $a, b \in Field_p$. Users and devices choose a private key PrK_a and then compute the public key $PuK_a = PK_a \cdot Gen$. Preshared symmetric keys are also generated and shared between users and gateways

5) REGISTRATION PHASE:

During this phase, the user will be registering with the gateway, during which it provides a newly captured image of the fingerprint and raw accelerometer data and also chooses a unique identity for the user. Then the data which will be transmitted is encrypted using a preshared key. Once the gateway receives the message, it performs image analysis and liveness detection on the fingerprint image to prevent spoofed fingerprints and reuse old fingerprints, and the model training is initiated for sensor fingerprinting after the features are extracted from the raw accelerometer data, which is stored for future validations. A secret is generated with a freshly generated nonce and is split into two shares using a random number. After a successful run of the above, the shares which are generated are stored, necessary details are shared with the user again. Detailed step by step process is as described in the fig. 1

6) AUTHENTICATION AND KEY ESTABLISHMENT:

This is triggered when user tries to access a device. First, the user capture an image of the fingerprint, sensor data is captured in the background by the application. The gateway ensures the image and accelerometer data are fresh by validating the metadata. On successful validation of the metadata, the liveness of the finger in the image is validated, biometrics are matched and sensor data is sent into the model for identification. After validation, the gateway retrieves the hash of the secrets and shared to the user and device with shares. User, Device use this information to establish the session key. The Steps are as shown in the fig. 2

7) RE-REGISTRATION PHASE:

Initialised when user wants to update the biometrics or add a new device by removing the old device. The steps are as mentioned below.

Step 1: user is prompted to take an image of his finger, and accelerometer data is captured in the background. Smartphone generates a nonce UN_i and computes $C_i = H(RREG, TID_i, TID_g, UN_i, H(User_{image}), H(User_{accel}), PS_{key})$ where $TID_i = H(UID_i, PuK_{user}, UN_i)$ and $TID_g = H(UID_g, PuK_{gateway}, UN_i)$. user device sends $RREG, TID_i, TID_g, H(User_{image}), H(User_{accel}), UN_i, C_i, \{User_{image}\}PS_{key}, \{User_{accel}\}PS_{key}$ to the gateway
Step 2: Gateway verifies UN_i and C_i validates the integrity of the message received. Gateway decrypts $User_{image}$, $User_{accel}$ and verifies their integrity using the hashes. After this, steps 2 and 3 are done from the user registration process, as shown in fig. 1 for registration of this new device, followed by the authentication and key establishment phase to access the device in smart homes.

VI. SECURITY ANALYSIS

The security analysis of the framework has already been performed in [16] which shows that the framework is robust and also resilient to many attacks like fake gateways, man-in-the-middle, replay attacks, etc. In this we have done the security analysis on the models which are introduced into the protocol to show that the resiliency of the model has not been

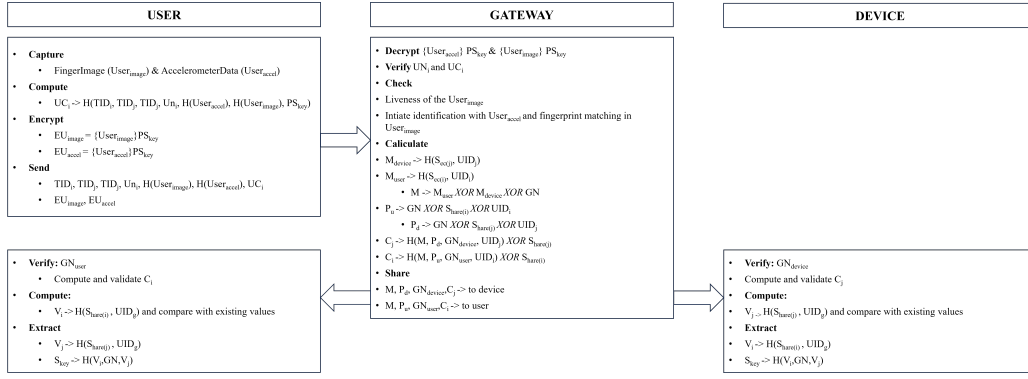


Fig. 2. Authentication and key establishment

	CPU Usage (%)	Memory Usage (MB)	Training Time (min)	Inference Time (Sec)
Biometrics	24 - 54	100 - 200	N/A	0.7
SensorFingerprinting	50 - 100	200 - 500	15	240

Fig. 3. Computation Needs in Raspberry PI

affected due to machine learning models. Goodfellow et al. proposed Fast Gradient Sign Method (FGSM) in [7] and [4] show the sensitivity of models to small perturbations, which gives the idea of testing the models under adversarial perturbations. Cohen et al. [11] proposed a framework for obtaining certified robustness through randomised smoothing, and Yuan et al. provides context and references to empirical and formal evaluation approaches in adversarial machine learning.

We have performed similar tests to check the performance and behaviour of the model to various noises which can affect the model.

A. Liveness Detection Model

The graphs below show the robustness of the liveness detection machine learning model. The ROC curve in Fig. 4 shows the effectiveness of the model without any false positives or false negatives. The reason for this perfect graph is the use of a dataset that is specific for spoof and live biometrics and the large amount of data available. Fig. 5 shows the accuracy of the model when the input is induced with Gaussian noise and FGSM accuracy ; both have identical accuracy. whereas the Avg. Certified Radius shows steady growth suggesting models robustness is improving with stronger perturbations.

Randomised smoothing is a technique used to certify the robustness of the machine learning model. The graphs in Fig ?? show that the model is robust and can handle up to 1.85 (approx.) confidently across all the samples.

B. Sensor Fingerprinting Models

Similar to above, analysis has been done for the random forest and SVM also, and the output graphs are shown as below in Fig. 6 We can see the effect of noise on the models, and from the graphs we can infer that both the models are robust and resilient to perturbations, making the models trustworthy.

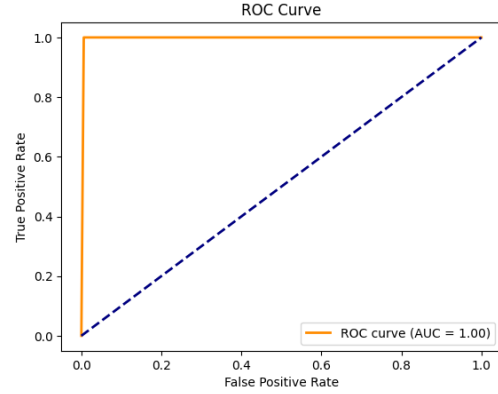


Fig. 4. ROC Curve

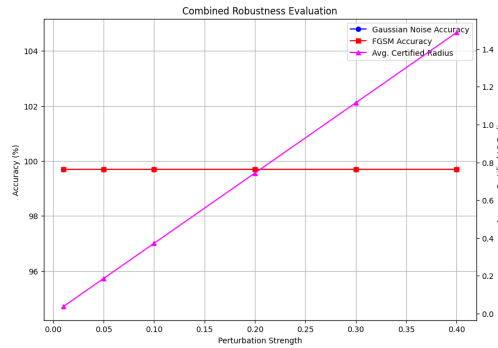


Fig. 5. Consolidated Graph for Liveness Detection

VII. CONCLUSION

We provide an efficient security protocol which combines biometrics and sensor fingerprinting, which is done using machine learning models, geometric secret sharing for authentication and secure communication. The analysis shows that the

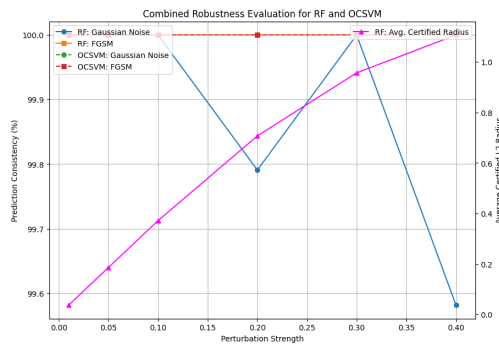


Fig. 6. Random Forest and SVM Consolidate graph

models are robust and resilient to perturbations. This simple implementation can go a long way in mutual authentication with the use of machine learning on low-powered devices. We have tried running these machine learning training and inference on a raspberryPi to check the resource requirements and found as shown in the table 3 To improve this further, we can use or train models with continuous learning, which can adapt to new data if needed to avoid retraining the model from the start and adjust the pipeline to reduce the load on the IOT device.

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